

CHENHAOYUE·WANG

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EDUCATION

University of California, Los Angeles (UCLA)

Sep 2021 - Jun 2026 (expected)

Ph.D. Candidate, Materials Science & Engineering

Areas of Expertise: Computational materials physics, electronic transport, and machine learning for quantum materials.

Academics: Cumulative GPA: 3.88/4.0

Soochow University

Sep 2015 - Jul 2019

Bachelor of Engineering, New Energy Materials and Devices (Highest Honor)

Areas of Expertise: Lithium-ion battery

Academics: Cumulative GPA: 3.7/4.0; Major GPA: 3.8/4.0; Ranking: 4/120

PUBLICATIONS

- **Chenhaoyue Wang**, Amartya S. Banerjee. Mechanical and Electronic properties of novel flat band material: buckled Silicon Kagome (2025). [In Preparation]
- **Chenhaoyue Wang**, Carlos Garcia-Cervera, Amartya S. Banerjee. Bloch Oscillations and Landau-Zener Transitions in Flat-Band Lattices with Quadratic and Linear Band Touchings (2025). [Available at: <https://doi.org/10.48550/arXiv.2510.03530>]
- Shivam Sharma, **Chenhaoyue Wang**, Hsuan Ming Yu, and Amartya S. Banerjee, Electronic States and Mechanical Behaviors of Phosphorus Carbide Nanotubes – Structural and Quantum Phase Transitions in a Quasi-one-dimensional Material (2025). [Available at: <https://arxiv.org/abs/2501.11239>]
- Md Tusher Ahmed, **Chenhaoyue Wang**, Amartya S. Banerjee and Nikhil Chandra Admal. Bicrystallography-informed Frenkel–Kontorova model for interlayer dislocations in strained 2D heterostructures. *Mechanics of Materials*, 190, 104903 (2024).
- Md Tusher Ahmed, **Chenhaoyue Wang**, Amartya S. Banerjee and Nikhil Chandra Admal. Multiscale analysis of large twist ferroelectricity and swirling dislocations in bilayer hexagonal boron nitride (2025). [Available at: <https://doi.org/10.48550/arXiv.2510.01419>]

RESEARCH EXPERIENCE

Graduate Student Researcher, NbSe₂ Strain-Induced Phonon Instability and CDW

Sep 2025 – present

- Performed phonon dispersion (DFPT) calculations for 1T and 2H phases of NbSe₂ under strain engineering, captured soft-mode evolution and CDW suppression trends.
- Conducted large-scale molecular dynamics (MD) simulations for 2H-NbSe₂ containing localized 1T domain to explore the atomic-scale mechanism of CDW transition, correlating with experimental STM observations.

Graduate Student Researcher, Quantum and Spin Transport Behavior in 2D Materials

Jun 2025 – Present

- Built tight-binding Hamiltonians incorporating intrinsic (Kane-Mele) spin-orbit coupling to quantized spin Hall conductivity under in-plane electric fields, and quantified Rashba-induced reduction impact on it.
- Developed a time dependent evolution framework based on Kubo formula Non-Equilibrium Adiabatic Steady-State (NEASS) formalisms to study the evolution of spin conductivity of 2D system with both extrinsic(Rashba) and intrinsic (Kane-Mele) spin-orbit coupling involved.

Graduate Student Researcher, Machine Learning-Based Magnetism Property Prediction in Nanotube Materials

Mar 2025 – Present

- Compiled datasets of mechanical, electronic and magnetic descriptors across diverse nanotube chiralities and elements.
- Trained machine-learning regression models to predict magnetic behavior from structural parameters such as strain, defect concentration, and chirality, enabling structure-to-magnetism mapping in quasi-1D systems.

Graduate Student Researcher, Mechanical and Electronic properties of novel flat band material: buckled Silicon Kagome (SiKL)

Nov 2021 – Jun 2025

- Computing band structures and phonon dispersion of silicon Kagome lattice using real-space density functional theory (DFT) calculations with SPARC and Quantum Espresso.
- Designing dynamically stable Kagome-like silicon structures by applying phonon-eigenvector perturbations to eliminate imaginary modes and reveal metastable buckled phases.

- Performed large-scale MD simulations (LAMMPS) for systems under different initial strains for long time range to investigate their thermal stability and possible phase evolution.
- Quantified crystal degree of system by radial distribution function (RDF) and constructed strain-temperature phase diagram based on error fitting function.
- Identified thermal stability up to 600K after strain improvement and determined a critical strain (~2%) separating disorder accumulation behavior and mechanical performance, which corresponding to the bonding environment shift.
- Modeled sample's growth on multilayer substrate Ag(111) with DFT and observed a natural strain (~0.1%) induced according interlayer correlation of silicon with silver, and demonstrated thermal stability up to 446 K with preserved flat-band behavior on large scale.
- Built automated Python workflows (NumPy, ASE, SciPy) for data processing, RDF analysis, bonding environment analysis, and energy tracing; employed machine learning regression to predict strain-dependent electronic features.

Graduate Student Researcher, Bloch Oscillations and Landau-Zener Transitions in Flat-Band Lattices with Quadratic and Linear Band Touchings

Sep 2023 – Mar 2025

- Developed a coherent-transport framework to simulate Bloch oscillations (BOs) in 2D materials under uniform electric fields using symplectic ODE solvers.
- Observed irregular BOs triggered by Landau-Zener tunneling (LZT) between dispersive and flat bands in Kagome and Lieb lattices.
- Derived effective two- and three-level Hamiltonians via tight-binding expansions, establishing analytical scattering-matrix models to quantify tunneling probabilities and phase accumulation.
- Demonstrated how strain-induced topology transitions from Kagome (quadratic touching) to Lieb (linear crossings) lattices modify interband coupling and irregular BO frequencies.

Graduate Student Researcher, Multiscale analysis of large twist ferroelectricity and swirling dislocations in bilayer hexagonal boron nitride

Jun 2024 – Mar 2025

- Performed DFT and Berry-phase polarization (in-plane and out-of-plane) calculations (ABINIT) for twisted bilayer hBN and graphene using PBE-D3, meta-GGA, and vdW-DF functionals.
- Demonstrated interlayer translation dependence of polarization inversion relevant to energy-storage and tunneling devices.

Graduate Student Researcher, Electronic States and Mechanical Behaviors of Phosphorus Carbide Nanotubes – Structural and Quantum Phase Transitions in a Quasi-one-dimensional Material

Dec 2023 – Oct 2024

- Performed ab initio molecular dynamics (AIMD) simulations using Quantum ESPRESSO (QE) and SPARC Cyclix to evaluate chirality-dependent stability in phosphorus-carbon (P_2C_3) nanotubes.
- Investigated the stability of flat bands and localized electron states near the Fermi level under elastic deformations.
- Analyzed the impact of hydrogen doping and vacancy under radial strain engineering on the magnetic properties of P_2C_3 nanotubes using density functional theory (DFT) calculations.

Graduate Student Researcher, Stacking engineering of Twisted van der Waals materials

Jul 2022 – Nov 2023

- Computing generalized stacking-fault energy for large-twist-angle bilayer graphene and via density functional theory (DFT).
- Analyzing twist-induced in-plane and out of plane polarization and established relationships between twist angle, interlayer dislocation, and ferroelectric domain formation.

RESEARCH EXPERIENCE – Battery Experiments

Research Assistant, Size Engineering of Graphene Shells by CVD Method

Sep 2018 - Jan 2020

- Fabricated graphene shells of controllable sizes via chemical vapor deposition (CVD). Explored the synergy effects of the size of graphene shells on its morphology properties and electrochemical performance.
- Discovered the empirical rule of graphene shell growth. Realized the specific capacity of graphene shells showed a bell-curve mode at a high current density as a function of graphene shell size. Designed optimal graphene shells with outstanding structural stability and electronic capacity.
- Computed the transmission and absorption spectra of graphene-shell based metamaterial in different arrays with COMSOL to investigate the photoelectric performances of graphene shells.

Project Leader, Preparation of Graphene Shell and Its Application in Lithium-ion batteries

Jun 2016 - Mar 2019

- Utilized Scanning Electron Microscope (SEM), Energy-Disperse X-rays (EDX), Raman Spectroscopy and Thermal Analysis to characterize graphene shells synthesized by CVD under different conditions. Explored and understood electrochemical performance and the lithium storage mechanism of graphene shells as active anode materials in Li-ion batteries.
- Optimized electrode materials and achieved a 25% increase in specific capacity of graphene shells at high current density

enhanced the cycling ability. Kept specific capacity above 90% of the origin after 500 cycles at the current density of 0.1C.

- Selected as National Program of Innovation and Entrepreneurship for Undergraduates and received 10000 RMB research funding. Invited to give a poster session in Soochow University Academic Conference.
- Investigating the optical power tuning conductivity and refractive index (RI) characteristics of multilayer-graphene shells in the visible optical frequency range.

TECHNICAL SKILLS

- Simulation & Modeling: DFT (Quantum ESPRESSO, SPARC, ABINIT), MD (LAMMPS) for diffusion, defect energy, and thermal stability, COMSOL Multiphysics for electro-thermal and mechanical modeling.
- Programming & Data: Python (NumPy, SciPy, Matplotlib, ASE, Numba, Ray, Scikit-learn) for data analysis, MATLAB and workflow automation, visualization (VESTA, Ovito), large-scale HPC runs (MPI and SLURM).
- Machine learning: Data pre-processing, feature extraction, regression modeling for property prediction.
- Battery & Experimental: Li-ion electrode fabrication, battery design, battery assembly, CVD synthesis, electrochemical testing (EIS, CV, charge-discharge cycling), SEM, Raman, glovebox operation.

CONFERENCE

- *Ab Initio Investigations of Kagome Xenos*, 2023 American Physical Society March Meeting
- *Ab Initio Investigations of Kagome Xenos*, 17th U. S. National Congress on Computational Mechanics, 2023
- *Ab Initio Investigations of a Two Dimensional form of Silicon featuring Dispersionless Electronic States*, 2024 Materials Research Society Spring Meeting
- *On Landau-Zener Transitions in Systems with Quadratic band Touching*, 2025 American Physical Society Meeting
- *Ab Initio Investigations of Buckled Silicon Kagome*, 18th U.S. National Congress on Computational Mechanics, 2025

HONORS & AWARDS

Nanomaterials 2025 Travel Award	2025
Outstanding Graduate	2019
Second Prize in “Internet Plus” Innovation and Entrepreneurship Competition	2019
Excellent Student Cadre	2018
Honorable Mention in 2018 Interdisciplinary Contest in Modeling	2018
Grand Prize for Innovation and Entrepreneurship, Soochow University	2018
Comprehensive Award, Soochow University	2018, 2017, 2016
Merit Student Award, Soochow University	2017
Award for Outstanding Social Work, Soochow University	2017, 2016
First Prize for Excellent Academic Performance, Soochow University	2018, 2016
Community Contribution Award, Soochow University	2017